Lección 10

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Plan de trabajo

Formulación Lagrangiana

– Aplicaciones: movimiento en campo central, dispersión de partículas

  * **Problema inverso de dispersión**
  * Observadores no inerciales
  * Movimiento de Sólido rígido

► Nuestro Syllabus:

Nuestro Syllabus:

FIS-310:

UNIDAD 2. Formulación Lagrangiana
2.1. Variables Generalizadas
2.2. Clasificación de los Sistemas Mecánicos
2.3. Elementos de Cálculo Variacional
2.4. Ecuaciones de Lagrange
2.5. Aplicaciones: movimiento en un campo central, observadores no inerciales

FIS-345:

Unidad I: Principios Variacionales, Lagrangianos y Hamiltonianos

- Principio de Hamilton y Lagrangianos
- Simetrías y Leyes de Conservación (Teorema de Noether)

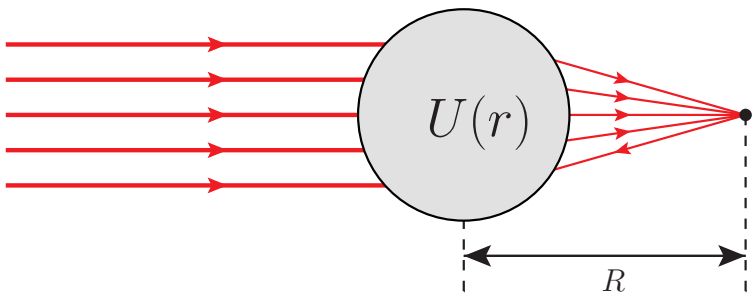
Unidad II: Algunas Aplicaciones

- Movimiento en campo de Fuerzas Centrales
- Movimiento de Sólido Rígido

*Inverse problem of scattering (formulation)

Sometimes instead of reconstruction of trajectory or evaluation of the cross-section in known potential, we may need to do the inverse: find the unknown interaction potential between particles from experimentally measured $d\sigma/d\Omega$, or our expectations about the trajectories (focusing potential)

(Landau, Section §18, Problem 7, *problem of Firsov*)



In the central field, the cross-section is directly related to $\theta(\rho)$:

$$\frac{d\sigma}{d\theta} = \pi \left| \frac{d\,\rho^2(\theta)}{d\theta} \right| \quad \Rightarrow \int_\theta^\pi \frac{d\sigma}{d\theta} = \pi\rho^2(\theta)$$

$$\theta = \pi - 2\int_{r_{\min}}^{\infty} \frac{\frac{v\rho}{r^2}\,dr}{\sqrt{\frac{2}{m}\left(\frac{mv_\infty^2}{2}\left(1-\frac{\rho^2}{r^2}\right) - U(r)\right)}}$$

$\Rightarrow$We always end up with the same problem:
$-$need to reconstruct $U(r)$ from known $\rho(\theta)$

$-$**Firsov's solution** ($E = \mathrm{const}$, known $\theta(\rho)$, unknown $w \equiv \sqrt{1-\frac{U}{E}}$):

$$w = \exp\left(-\frac{1}{\pi}\int_{rw}^{\infty} \cosh^{-1}\left(\frac{\rho}{rw}\right)\frac{d\theta(\rho)}{d\rho}\,d\rho\right) = \exp\left(\frac{1}{\pi}\int_{rw}^{\infty}\frac{\theta(\rho)}{\sqrt{\rho^2 - r^2w^2}}\,d\rho\right),$$

Example

Let's assume ($ad\text{-}hoc$)

$$\theta = \Theta(a - \rho) \arcsin\left(\frac{\rho}{a}\right)$$

i.e. scatters (has nonzero scattering angle) only if $\rho < a$, so

$$\rho(\theta) = a \sin\theta,$$

and the cross-section

$$\frac{d\sigma}{d\theta} \qquad = 2\pi\, \rho(\theta) \left|\frac{d\rho}{d\theta}\right| = \pi a^2 \sin 2\theta$$

Find the potential $U(r)$ which corresponds to this dependence

(we'll see a bit later practical application of this result)

Solution:

$$w \qquad = \exp\left(\frac{1}{\pi}\int_{rw}^{\infty}\frac{\theta(\rho)}{\sqrt{\rho^2 - r^2w^2}}\,d\rho\right)$$

$\Rightarrow$algebraic equation, allows to get $w(r)$:

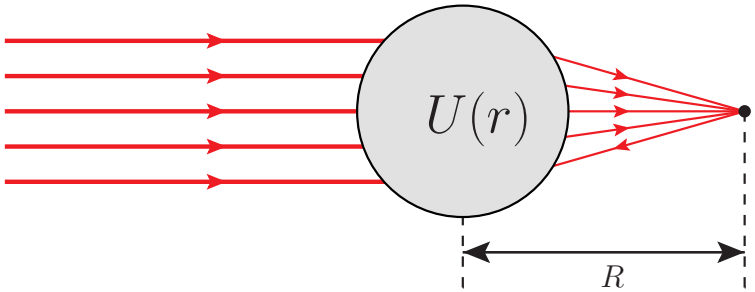
$$\frac{1}{\pi}\int_{rw}^{\infty}\frac{\theta(\rho)}{\sqrt{\rho^2 - r^2w^2}}\,d\rho = -\frac{1}{2}\ln\left(\frac{a}{\sqrt{a^2 - r^2w^2} + a}\right)$$

$$\Rightarrow w \equiv \sqrt{1 - \frac{U(r)}{E}} = \sqrt{2 - \frac{r^2}{a^2}}$$

$$U(r) = E \times \begin{cases} \left(\frac{r^2}{a^2} - 1\right), & r < a \\ 0, & r > a \end{cases}$$

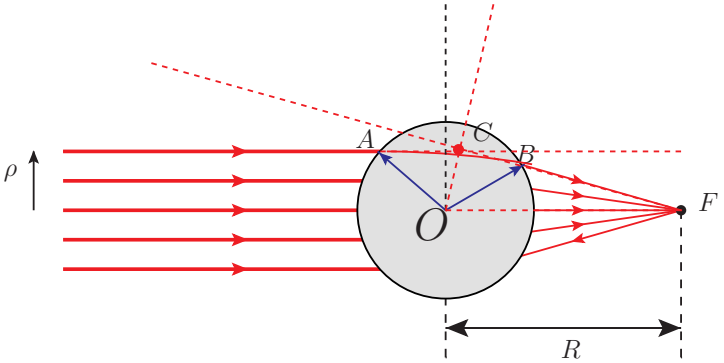
Focusing potentials

A collinear beam of monoenergetic particles with energy E and radius of the beam $\rho_{\max}$ scatters in the central potential $U(r)$. After the scattering all particles are focused to a point at large distance $R \gg \rho_{\max}$ from the center. Find the potential $U(r)$. [Luneburg lens configuration]



Different modifications/geometries are studied in the literature (*e.g.* need to focus particles from collinear beam, or emitted from point source, etc), here consider just one example

Focusing potentials



Solution:

►The impact parameter ρ is related to scattering angle θ as $\rho/R^* = \tan\theta$, where $R^* \approx R$ so

$$\rho \approx R\,\tan\theta,\ \ \theta \approx \arctan\left(\frac{\rho}{R}\right) \ll 1$$

More accurate analysis:

–Line OC forms angle $\theta/2$ with vertical (black) line; deflection angle θ corresponds to particle which passed a full path ACB

–Angles $\angle OCF = \angle COF = \pi/2 - \theta/2$, ⇒OFC is isosceles triangle, CF=OF=R

–Angles $\angle CFO = \theta$, so $\rho = R\sin\theta$

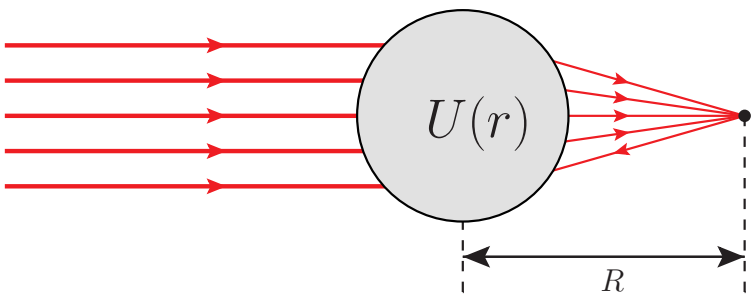
$$\theta = \arcsin\left(\frac{\rho}{R}\right) \quad \text{(exact)}$$

⇒We can use Firsov's solution to find $U(r)$

$$w \qquad = \exp\left(\frac{1}{\pi}\int_{rw}^{\rho_{\max}} \frac{\theta(\rho)}{\sqrt{\rho^2 - r^2 w^2}}\, d\rho\right)$$

►Formally integral diverges (or does not exist) if $\rho_{\max} = \infty$ ...

Focusing potentials



Solution:

►The impact parameter ρ is related to scattering angle θ as $\rho/R = \tan\theta$, so

$$\rho = R\tan\theta,\ \theta = \arctan\left(\frac{\rho}{R}\right)$$

⇒We can use Firsov's solution to find $U(r)$

$$w = \exp\left(\frac{1}{\pi}\int_{rw}^{\rho_{\max}} \frac{\theta(\rho)}{\sqrt{\rho^2 - r^2 w^2}}\, d\rho\right)$$

►Formally integral diverges if $\rho_{\max} = \infty$ ...

Comment on divergence at $\rho \to \infty$

$$\rho = R\,\tan\theta,\ \theta = \arctan\left(\frac{\rho}{R}\right)\ \text{or}\ \theta = \arctan\left(\frac{\rho}{\mathrm{R}}\right)$$

$$w \quad = \exp\left(\frac{1}{\pi}\int_{rw}^{\rho_{\max}} \frac{\theta(\rho)}{\sqrt{\rho^2 - r^2 w^2}}\, d\rho\right) \qquad (1)$$

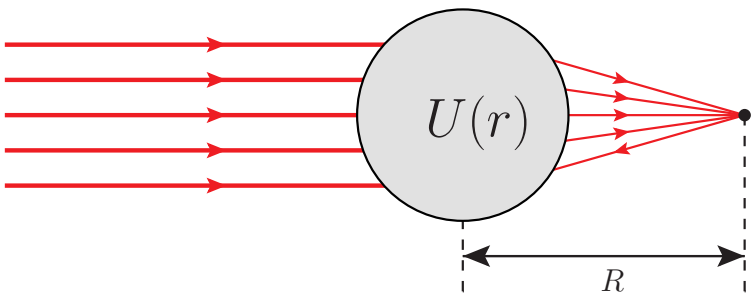
Formally integral diverges if $\rho_{\max} = \infty$. This happens because this potential requires that $\theta(\rho = \infty) \approx \pi/2$, *i.e.* even particles which go very far from center should scatter.
−Earlier in derivation of (1) we always assumed that $\theta(\rho \to \infty) = 0$.
−If we use alternative/original form of Firsov's formula

$$w \quad = \exp\left(-\frac{1}{\pi}\int_{rw}^{\infty} \cosh^{-1}\left(\frac{\rho}{rw}\right) \frac{d\theta(\rho)}{d\rho}\, d\rho\right), \qquad (2)$$

then can see that the divergence at $\rho \to \infty$ does not happen (as You may remember, we got (1) from (2) integrating by parts and assuming that $\theta(\infty) = 0$ to suppress the surface terms)

May finalize with Mathematica, solution is elementary function (though lengthy!)

Focusing potentials



►If we consider $\rho_{\max} \ll R$ (need to focus narrow bin), then $\theta \ll 1$,

$$\tan\theta \approx \sin\theta \approx \theta$$

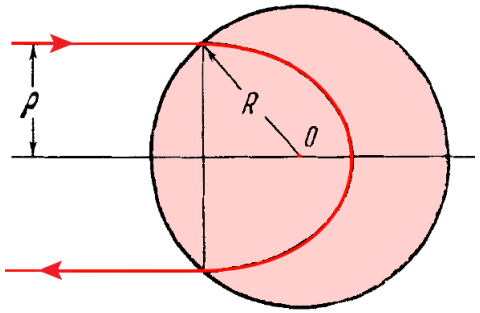
and we may use analytic result from previous problem,

$$U(r) = E \times \begin{cases} \left(\frac{r^2}{a^2} - 1\right), & r < a \\ 0, & r > a \end{cases}$$

►Our potential depends on energy (E)
–will focus (as lens) only the particles of a given energy

Example problem 3

The collinear beam of electrons of energy E_0 scatters in the central field $U(r)$. For the sake of simplicity, You may assume that the potential vanishes at distances $r \geq R$ from the center (point O). After the scattering, all the electrons with impact parameters $\rho < R$ are reflected back, as shown in the Figure.

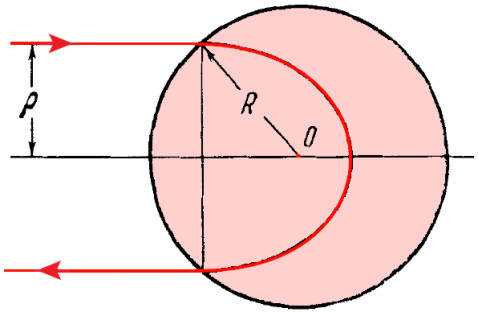


The red line with arrows is the trajectory of electrons, the colored area is the region $r < R$ where the central potential $U(r)$ is not zero.

1. Find the potential $U(r)$ which satisfies these properties.
2. A collinear beam of particles with energy $E \neq E_0$ scatters in the central potential $U(r)$ which You found earlier. Find the cross-section of elastic scattering $d\sigma/d\Omega$ for this case. Analyze what happens with the cross-section in the limit $E \to E_0$.

Goal of this problem:
Check your knowledge of inverse scattering motion in central field

Example problem 3



From geometric construction can see that the scattering angle

$$\theta(\rho < R) = \pi$$

and there is no scattering outside

This is classical inverse scattering problem,
– Since all particles have the same energy, can use Firsov's formula

$$w \qquad = \exp\left(\frac{1}{\pi}\int_{rw}^{R}\frac{\theta(\rho)}{\sqrt{\rho^2 - r^2w^2}}\,d\rho\right)$$

$$\ln w = \frac{1}{\pi}\int_{rw}^{R}\frac{\theta(\rho)}{\sqrt{\rho^2 - r^2w^2}}\,d\rho = \coth^{-1}\left(\frac{R}{\sqrt{R^2 - r^2w^2}}\right),$$

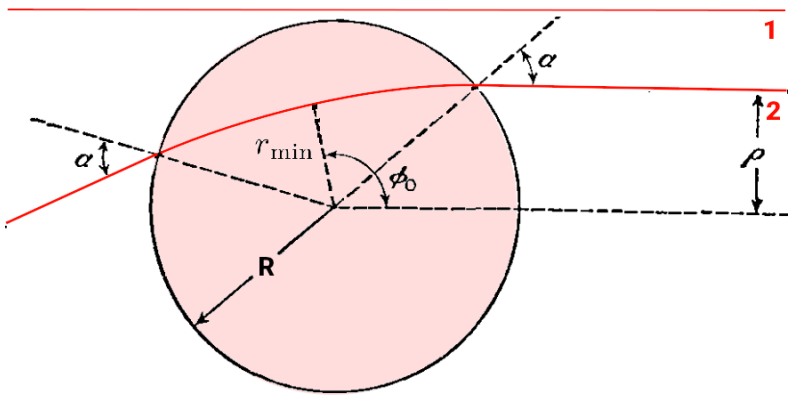
$$w = \sqrt{\frac{2R - r}{r}} = \sqrt{1 - \frac{U}{E_0}}$$

so

$$U(r < R) = 2E_0\left(1 - \frac{R}{r}\right), \quad U(r > R) = 0$$

Example problem 3

Different regimes:



Should distinguish:

- For $r > R$ trajectories are just straight lines
- For $r < R$ need to find trajectories in the potential

$$U(r) = \left(2E_0 - \frac{2E_0 R}{r}\right), \quad r < R, \quad r <$$

The first term $2E_0 = \mathrm{const}$ just "shifts" energy, does not contribute to force $\boldsymbol{F} = -\boldsymbol{\nabla} U$
The second term is ordinary Coulomb, need only to change $\alpha \to 2E_0 R$
$\Rightarrow$We may literally repeat what we discussed in Kepler's problem and get solutions as conic section

$$\frac{p}{r} = 1 + e\cos\phi, \quad p = \frac{M^2}{2mE_0 R}, \quad e = \sqrt{1 + \frac{2\left(E - 2E_0\right)M^2}{4mE_0^2 R^2}}$$

If particle's trajectory leaves the potential well, the trajectory is a mix of straight line AND conic section
– No discontinuity at the border since $U(r)$ is continuous

Example problem 3

$$\frac{p}{r} = 1 + e\cos\phi, \quad p = \frac{M^2}{2mE_0 R}, \quad e = \sqrt{1 + \frac{2\left(E - 2E_0\right)M^2}{m\alpha^2 R^2}}$$

NOTE: If You use Mathematica and try to get trajectory using general expression

$$\phi - \phi_0 = \int^r dr \, \frac{M}{m\, r^2 \sqrt{\frac{2}{m}\left(E - U_{\text{eff}}\left(r\right)\right)}}$$

You may get result in many equivalent forms:

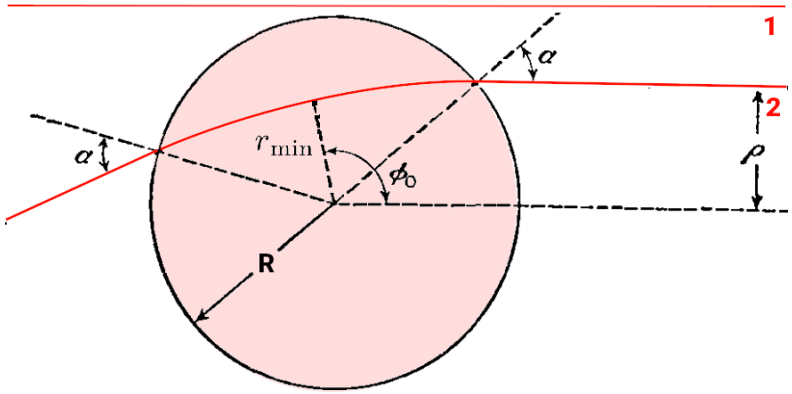
$$\phi = \arccos\left[\frac{1}{e}\left(\frac{p}{r} - 1\right)\right]$$

$$\phi = \arcsin\left[\sqrt{1 - \frac{1}{e^2}\left(\frac{p}{r} - 1\right)^2}\right]$$

$$\phi = \arctan\left[\frac{\sqrt{1 - \frac{1}{e^2}\left(\frac{p}{r} - 1\right)^2}}{\frac{1}{e}\left(\frac{p}{r} - 1\right)}\right] = \arctan\left[\frac{\sqrt{(e^2 - 1)r^2 + p^2 + 2pr}}{(p - r)}\right]$$

Example problem 3

Different regimes:



- for $\rho > R$ no scattering at all (!!!) $\Rightarrow$ $\theta = 0$, but $\phi_0 = \pi/2, \neq 0$
- For $\rho < R$ we may use conventional formulas valid in any central potential (show in blue color additional terms which differ from Kepler's problem):

$$\phi_0 = \int_{r_{\min}}^{\infty} \frac{\frac{M}{m\,r^2}\,dr}{\sqrt{\frac{2}{m}\left(E - U(r) - \frac{M^2}{2\,m\,r^2}\right)}} = \int_{r_{\min}}^{\infty} \frac{\frac{\rho}{r^2}\,dr}{\sqrt{\left(1 - \frac{4E_0\Theta(R-r)}{mv_\infty^2}\left(1 - \frac{R}{r}\right) - \frac{\rho^2}{r^2}\right)}}$$

$$= \int_{r_{\min}}^{R} \frac{\frac{\rho}{r^2}\,dr}{\sqrt{\left(1 - \frac{4E_0}{mv_\infty^2}\left(1 - \frac{R}{r}\right) - \frac{\rho^2}{r^2}\right)}} + \int_{R}^{\infty} \frac{\frac{\rho}{r^2}\,dr}{\sqrt{\left(1 - \frac{\rho^2}{r^2}\right)}}$$

Minimal distance $r_{\min}$ found from equation

$$1 - \frac{4E_0 U_0}{mv_\infty^2}\left(1 - \frac{R}{r_{\min}}\right) - \frac{\rho^2}{r_{\min}^2} = 0, \quad \rho \le R, \quad 0 \le r_{\min} \le R, \quad r_{\min} = \frac{p}{1+e}$$

Example problem 3

Evaluation of the integrals in Mathematica for $\phi_0 = (\pi - \theta)/2$:

$$\phi_0 = \sec^{-1}\left(\frac{R\sqrt{m\rho^2 v^2\left(mv^2 - 4E_0\right) + \alpha^2 R^2}}{m\rho^2 v^2 - 2E_0 R^2}\right) + \tan^{-1}\left(\frac{\rho}{\sqrt{R^2 - \rho^2}}\right)$$

To "join" several inverse functions, can apply known function (Sin, Cos, ...) use TrigExpand[...] and then FullSimplify[...], final results:

$$\cos\phi_0 = -2E_0\sqrt{\frac{R^2 - \rho^2}{m\rho^2 v_\infty^2\left(mv_\infty^2 - 4E_0\right) + 4E_0^2 R^2}}, \quad \rho = \frac{2E_0 R\sin(\phi_0)}{\sqrt{4E_0^2 + mv^2\cos^2(\phi_0)\left(mv^2 - 4E_0\right)}}$$

$$\sin\phi_0 = \frac{\rho\left(mv_\infty^2 - 2E_0\right)}{\sqrt{m\rho^2 v_\infty^2\left(mv_\infty^2 - 4E_0\right) + 4E_0^2 R^2}}, \quad \rho = \frac{2E_0 R\sin(\phi_0)}{\sqrt{\left(2E_0 - mv^2\right)^2 - mv^2\sin^2(\phi_0)\left(mv^2 - 4E\right.}}$$

$$\tan\phi_0 = \frac{\rho\left(2E_0 - mv^2\right)}{2E_0\sqrt{R^2 - \rho^2}}, \quad \rho = \frac{2E_0 R\tan(\phi_0)}{\sqrt{4E_0^2 + m^2 v^4 - 4E_0 mv^2 + 4E_0^2\tan^2(\phi_0)}}$$

All the expressions for ρ in right-hand sides are equivalent to each other
To find cross-section, can use

$$\frac{d\sigma}{d\Omega} = \frac{1}{2\pi\sin\theta}2\pi\rho\left|\frac{d\rho}{d\theta}\right| = \frac{1}{2\sin\theta}\left|\frac{d\rho^2}{d\theta}\right| = \frac{4E_0^2 R^2\left(2E_0 - mv^2\right)^2}{4\left(4E_0^2\cos^2\left(\frac{\theta}{2}\right) + \left(2E_0 - mv^2\right)^2\sin^2\left(\frac{\theta}{2}\right)\right)^2}$$

Example problem 3

$$U(r) = \begin{cases} 2E_0\left(1 - \frac{R}{r}\right), & r < R \\ 0, & r \geq R \end{cases} \qquad (1) \qquad \rho = \frac{2E_0 R\cot(\theta/2)}{\sqrt{4E_0^2 + m^2v^4 - 4E_0 mv^2 + 4E_0^2\cot^2(\theta/2)}}$$

$$\frac{d\sigma}{d\Omega} = \frac{E_0^2R^2\left(2E_0 - mv^2\right)^2}{\left(4E_0^2\cos^2\left(\frac{\theta}{2}\right) + (2E_0 - mv^2)^2\sin^2\left(\frac{\theta}{2}\right)\right)^2} = \frac{E_0^2R^2\left(E_0 - E\right)^2}{\left(E_0^2\cos^2\left(\frac{\theta}{2}\right) + (E_0 - E)^2\sin^2\left(\frac{\theta}{2}\right)\right)^2}$$

Observations and cross-check:

—If $E = mv^2/2$ approaches E_0, the cross-section vanishes (as expected)

—Our potential vanishes at large $r \to \infty$, so we don't have singularity at small scattering angles

—If we replace $E_0 \to E_0/R$ and after that consider limit $R \to \infty$, $U(r)$ turns into the Coulomb potential, and $d\sigma/d\Omega$ turns into Rutherford's formula

*Inverse problem of scattering: 2 other known solutions

Assume that we know from experiment the elastic cross-section $d\sigma/d\theta$ for particles of **fixed angular momentum** M as a function of energy E. Evaluate the potential $V(r)$.

(Problem of Hoyt)
Final result:

$$\Rightarrow \frac{\lambda}{r} = \sqrt{U} + \\ + \frac{\lambda^2}{\pi\sqrt{U}} \int_{\lambda/\sqrt{U}}^{\infty} \frac{\theta(\rho)d\rho}{\rho^2\sqrt{\rho^2 - \lambda^2/U}}$$

$$\lambda = M/\sqrt{2m}, \quad U = V + \lambda^2/r^2$$

Assume that we know from experiment the elastic cross-section $d\sigma/d\theta$ for particles of **fixed impact parameter** ρ as a function of energy E. Evaluate the potential $U(r)$.

(Bogdanov-Demkov's problem)
Final result:

$$\arcsin\left(\frac{\rho}{r(U)}\right) = \frac{1}{2\pi}\int_0^U \frac{\pi - \theta(E)}{\sqrt{UE - E^2}}$$

$$U = \frac{r^2}{r^2 - \rho^2} V(r)$$

Non-inertial RF: why it matters?

So far we always assumed that we work in the inertial frames
−Use Lagrangian or Newtonian formalism ...

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... but sometimes non-inertial frames are more natural/more convenient.

Now we need to recall everything we remember from undergraduate courses about the non-inertial reference frames ...

Control question I

Is it possible to use Newtonian formalism in the noninertial reference frames?

–If yes, what should we change?

Control question II

Is it possible to use Lagrangian formalism in the noninertial reference frames?

–If yes, what should we change?

Control question III

Please provide explicit expressions for all the fictitious forces which appear in the non-inertial frame

–Explain to derive these explicit expressions
  A) In Newtonian formalism
  B) In Lagrangian formalism

Control question IV

Assume that You want to demonstrate that the laboratory is a non-inertial frame due to the Earth's rotation. Please suggest a mechanical experiment which can show this.

Control question V

Does the non-inertial motion of the reference frame affect non-mechanical forces (e.g. electrostatic or magnetic?)

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–Use Lagrangian or Newtonian formalism ...

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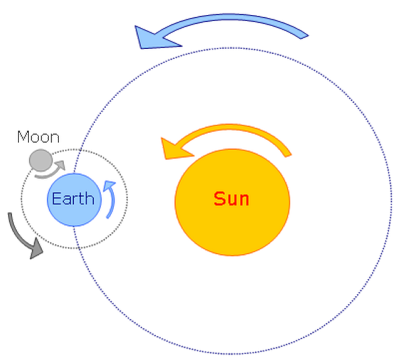
... but sometimes non-inertial frames are more natural/more convenient.

Example of non-inertial frame



Our laboratory from the point of view of remote observer:

–rotates with period $T_1 \approx 1$ day around the Earth's axis

–rotates with period $T_2 \approx 1$ year around the Sun.

⇒**The laboratory on the Earth is not inertial, the rotations will affect trajectories (if precision is sufficient).**

If the Earth were at rest:

–*Kepler's laws: Trajectory of the Earth's satellite (and the Moon) are plane ellipses*

***How the trajectory will change due to the orbital motion of the Earth?**

(effect is tiny ... frequently masked by other perturbations)

Non-inertial RF: what changes?

Assume that:

- $\boldsymbol{e}_a \equiv \{\hat{i}, \hat{j}, \hat{k}\}$ are Cartesian basis vectors of the **inertial** reference frame K. Each of them is constant,

$$\frac{d\boldsymbol{e}_a}{dt} = 0.$$

- $\boldsymbol{e}'_a \equiv \{\hat{i}', \hat{j}', \hat{k}'\}$ are Cartesian basis vectors of the **non-inertial** reference frame K'. Each of them is not constant due to motion of the reference frame K',

$$\frac{d\boldsymbol{e}'_a}{dt} \neq 0.$$

- We can decompose arbitrary vector $\boldsymbol{A}$ in both referece frames,

$$\begin{aligned} \boldsymbol{A} &= \sum A_a \boldsymbol{e}_a = \sum A'_a \boldsymbol{e}'_a, \\ A_a &= (\boldsymbol{e}_a \cdot \boldsymbol{A}), \qquad A'_a = \left(\boldsymbol{e}'_a \cdot \boldsymbol{A}\right). \end{aligned}$$

$\Rightarrow$if vector $\boldsymbol{A}$ is constant, its Cartesian components A_a are constant in inertial reference frame K, yet its components are NOT constant in non-inertial frame K' due to motion of basis vectors.

Non-inertial RF: what changes?

- We can decompose arbitrary vector $\boldsymbol{A}$ in both reference frames,

$$\begin{aligned} \boldsymbol{A} &= \sum A_a \boldsymbol{e}_a = \sum A'_a \boldsymbol{e}'_a, \\ A_a &= (\boldsymbol{e}_a \cdot \boldsymbol{A}), \qquad A'_a = \left(\boldsymbol{e}'_a \cdot \boldsymbol{A}\right). \end{aligned}$$

$\Rightarrow$if vector $\boldsymbol{A}$ is constant, its Cartesian components A_a are constant in inertial reference frame K, yet its components are NOT constant in non-inertial frame K' due to motion of basis vectors.

$\Rightarrow$If there is no forces acting on particle, its velocity vector $\boldsymbol{v}$ (all its inertial frame components) should be constant in the inertial frame K. However, in non-inertial frame K' its components are NOT constant !!!

$$\Downarrow$$

The equations of motion should change their form in non-inertial frame

Equations of motion in non-inertial system

Lagrange approach is covariant under substitution of coordinates

$$\{x, y, z\} \to \{q_1, q_2, q_3\}$$

$\Rightarrow$we may start from inertial frame, replace coordinates with non-inertial and write out the corresponding Euler-Lagrange equations

Equations of motion in non-inertial system

Lagrangian & EOM in inertial RF

$$L = \frac{m}{2}\dot{\boldsymbol{r}}^2 - U(\boldsymbol{r}), \qquad m\ddot{\boldsymbol{r}} = -\left(\frac{\partial U}{\partial \boldsymbol{r}}\right),$$

Assume that the reference frame K' moves with velocity $\boldsymbol{V} = \boldsymbol{V}(t)$ and rotates with constant angular velocity $\boldsymbol{\Omega}$ with respect to some inertial frame K, so the velocity of particle $\boldsymbol{v}$ is related to the velocity in inertial frame $\boldsymbol{v}_{\text{inertial}}$ as

$$\boldsymbol{v}_{\text{inertial}} = \boldsymbol{v} + \boldsymbol{\Omega} \times \boldsymbol{r} + \boldsymbol{V}(t).$$

Evaluate the lagrangian in the non-inertial reference frame and write out the corresponding equations of motion.

- Your lagrangian L might depend on the particle's radius-vector $\boldsymbol{r}(t)$ and its velocity $\boldsymbol{v}(t)$ in the non-inertial rest frame.
- The vectors $\boldsymbol{V}(t)$, $\boldsymbol{\Omega}(t)$ are NOT degrees of freedom, they are just fixed (known) functions

Equations of motion in non-inertial system

$$L = \frac{m}{2}\left(v^2 + \underbrace{(\boldsymbol{\Omega} \times \boldsymbol{r})^2}_{\text{centrifugal force}} + \underbrace{V^2(t)}_{\text{doesn't change EOM}} + \underbrace{2\boldsymbol{v} \cdot (\boldsymbol{\Omega} \times \boldsymbol{r})}_{\text{Coriolis}} + \underbrace{2\boldsymbol{v} \cdot \boldsymbol{V}(t)}_{\text{inertia}} \right) - U(r)$$

►Note that we can rewrite the term $\sim \vec{\boldsymbol{v}} \cdot \vec{\boldsymbol{V}}$ as

$$\vec{v} \cdot \vec{V} = -\vec{r} \cdot \frac{d\vec{V}}{dt} + \frac{d\left(\vec{r} \cdot \vec{V}\right)}{dt}$$

and we know that adding <u>full</u> derivative $d(...)/dt$ to Lagrangian does not affect the equations of motion.

$$\begin{aligned}
\frac{\partial L}{\partial \dot{r}_a} \equiv \frac{\partial L}{\partial v_a} &= m v_a + m\,(\boldsymbol{\Omega} \times \boldsymbol{r})_a + m\,V_a, \\
\frac{\partial L}{\partial r_a} &= -m\left[\vec{\Omega} \times [\boldsymbol{\Omega} \times \boldsymbol{r}]\right]_a + m\left[\vec{\boldsymbol{v}} \times \vec{\Omega}\right]_a + -\frac{\partial U}{\partial r_a} \\
\Rightarrow m\dot{\vec{v}} &= -\underbrace{m\dot{\vec{\Omega}} \times \vec{r}}_{\text{Euler force}} - \underbrace{m\vec{\Omega} \times [\vec{\Omega} \times \vec{r}]}_{\text{centrifugal}} - 2\underbrace{m\vec{\Omega} \times \vec{\boldsymbol{v}}}_{\text{Coriolis}} \underbrace{-m\dot{\vec{V}}}_{\text{inertia}} - \frac{\partial U}{\partial \vec{r}}
\end{aligned}$$

Equations of motion in non-inertial system

$$\Rightarrow m\dot{\vec{v}} = -\underbrace{m\dot{\vec{\Omega}} \times \vec{r}}_{\text{Euler force}} - \underbrace{m\vec{\Omega} \times [\vec{\Omega} \times \vec{r}]}_{\text{centrifugal}} - 2\underbrace{m\vec{\Omega} \times \vec{\boldsymbol{v}}}_{\text{Coriolis}} \underbrace{-m\dot{\vec{V}}}_{\text{inertia}} - \frac{\partial U}{\partial \vec{r}}$$

Equations of Newton

- Explain how to get the above-mentioned equations from the second law of Newton (not using the the formalism of Lagrange)

Derivation from EOMs in inertial RF

- In the moving frame the basis vectors $\vec{\boldsymbol{e}}_i$ are not constants, they rotate with angular velocity $\boldsymbol{\Omega}$:

$$\begin{aligned} \vec{\boldsymbol{e}}_i(t+\delta t) &= \vec{\boldsymbol{e}}_i(t) + \delta\vec{\boldsymbol{\varphi}}(t) \times \vec{\boldsymbol{e}}_i(t),\ \delta\vec{\boldsymbol{\varphi}} \ll 1 \\ \frac{d\vec{\boldsymbol{\varphi}}}{dt} &\equiv \vec{\boldsymbol{\Omega}} \Rightarrow \frac{d\vec{\boldsymbol{e}}_i}{dt} = \vec{\boldsymbol{\Omega}} \times \vec{\boldsymbol{e}}_i \end{aligned}$$

- Furthermore, due to relative motion of RF, components of radius-vector

$$\vec{\boldsymbol{r}} \equiv \sum_i r_i \vec{\boldsymbol{e}}_i,$$

and velocity $\boldsymbol{v}$ will also change,

$$\vec{\boldsymbol{v}}_{\text{inert}} = \vec{\boldsymbol{v}} + \vec{\boldsymbol{\Omega}} \times \vec{\boldsymbol{r}} + \vec{\boldsymbol{V}}$$

$$\begin{aligned} \vec{\boldsymbol{a}}_{\text{inert}} &= \left(\vec{\boldsymbol{a}} + \vec{\boldsymbol{\Omega}} \times (\vec{\boldsymbol{v}} + \vec{\boldsymbol{\Omega}} \times \vec{\boldsymbol{r}})\right) + \left(\dot{\vec{\boldsymbol{\Omega}}} \times \vec{\boldsymbol{r}} + \vec{\boldsymbol{\Omega}} \times \vec{\boldsymbol{v}}\right) + \dot{\vec{\boldsymbol{V}}} \\ &= \vec{\boldsymbol{a}} + \dot{\vec{\boldsymbol{\Omega}}} \times \vec{\boldsymbol{r}} + 2\vec{\boldsymbol{\Omega}} \times \vec{\boldsymbol{v}} + \vec{\boldsymbol{\Omega}} \times (\vec{\boldsymbol{\Omega}} \times \vec{\boldsymbol{r}}) + \dot{\vec{\boldsymbol{V}}} \qquad (1) \end{aligned}$$

Derivation from EOMs in inertial RF

$$\vec{\boldsymbol{v}}_{\text{inert}} = \vec{\boldsymbol{v}} + \vec{\boldsymbol{\Omega}} \times \vec{\boldsymbol{r}} + \vec{\boldsymbol{V}}$$

$$\begin{aligned}
\vec{\boldsymbol{a}}_{\text{inert}} &= \left(\vec{\boldsymbol{a}} + \vec{\boldsymbol{\Omega}} \times (\vec{\boldsymbol{v}} + \vec{\boldsymbol{\Omega}} \times \vec{\boldsymbol{r}})\right) + \left(\dot{\vec{\boldsymbol{\Omega}}} \times \vec{\boldsymbol{r}} + \vec{\boldsymbol{\Omega}} \times \vec{\boldsymbol{v}}\right) + \dot{\vec{\boldsymbol{V}}} \\
&= \vec{\boldsymbol{a}} + \dot{\vec{\boldsymbol{\Omega}}} \times \vec{\boldsymbol{r}} + 2\vec{\boldsymbol{\Omega}} \times \vec{\boldsymbol{v}} + \vec{\boldsymbol{\Omega}} \times (\vec{\boldsymbol{\Omega}} \times \vec{\boldsymbol{r}}) + \dot{\vec{\boldsymbol{V}}} \qquad (1)
\end{aligned}$$

$\Rightarrow$The second law of Newton $m\,\vec{\boldsymbol{a}}_{\text{inert}} = \vec{\boldsymbol{F}} = -\nabla U$, after substitution of (1), in non-inertial frame we'll get

$$\Rightarrow m\dot{\vec{v}} = -\underbrace{m\dot{\vec{\Omega}} \times \vec{r}}_{\text{Euler force}} - \underbrace{m\vec{\Omega} \times [\vec{\Omega} \times \vec{r}]}_{\text{centrifugal}} - 2\underbrace{m\vec{\Omega} \times \vec{\boldsymbol{v}}}_{\text{Coriolis}} \underbrace{-m\dot{\vec{V}}}_{\text{inertia}} - \frac{\partial U}{\partial \vec{r}} \qquad (2)$$

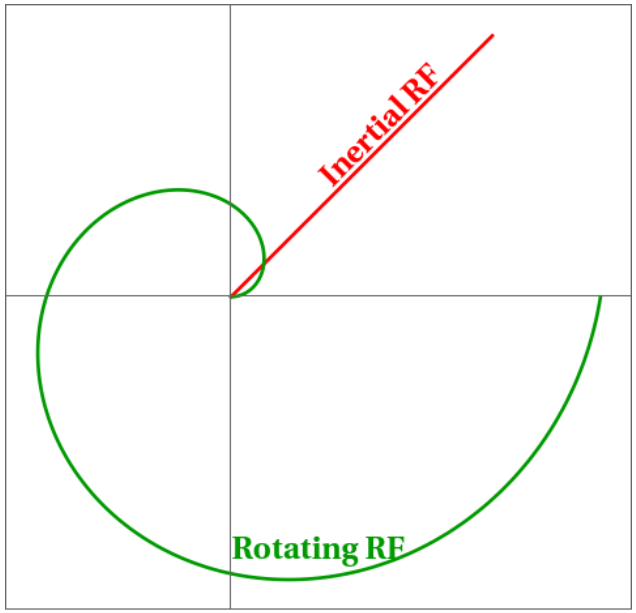
in agreement with what we found using Lagrangian approach.

Consider a mass sliding without friction from the center (1) of rotating turntable outside with velocity $\vec{v}_0$.

We expect that:

- In the lab frame, the trajectory is a straight line, *i.e.* $\{x(t), y(t)\}$ and $y(x)$ are all linear functions of their arguments
- In the rotating frame, the trajectory is a curved green line due to effects of additional fictitious forces

Expectation:



For simplicity assume that the particle slides from the center $(x, y) = 0$ outwards

−In lab frame

$$x = v_x t, \quad y = v_y t$$

−For observer in rotating frame

$$x = v_x t \cos \omega t, \quad y = v_y t \sin \omega t$$

► Now we will show explicitly that in absence of "true" external forces, the fictitious forces in rotating frame "bend" the trajectory in such a way that it remains a straight line in inertial ref. frame (IRF):

► Equation of motion:

$$m\,\ddot{\boldsymbol{r}} = -\underbrace{m\,\boldsymbol{\Omega}\times[\boldsymbol{\Omega}\times\boldsymbol{r}]}_{\text{centrifugial}} + \underbrace{2m\dot{\boldsymbol{r}}\times\boldsymbol{\Omega}}_{\text{Coriolis}}$$

► Assume that $\hat{z}$ point in direction of $\boldsymbol{\Omega}$ (perpendicular to the surface of the disk)

▷Motion occurs in x, y plane

$$\boldsymbol{\Omega}\times[\boldsymbol{\Omega}\times\boldsymbol{r}] = \underbrace{\boldsymbol{\Omega}(\Omega\cdot r)}_{=0} - \boldsymbol{r}\,\Omega^2$$

$$\ddot{x} \qquad = 2\,\dot{y}\,\Omega_z + \Omega_z^2 x, \quad \ddot{y} = -2\,\dot{x}\Omega_z + \Omega_z^2 y,$$

-homogeneous linear system of 2 ODEs, look for solutions $\{x, y\} \sim \{x_0,\, y_0\}e^{i\omega t}$. Can get that $\omega_{1,2} = \pm\Omega$ is a double root, so the solution has a form (*check with Mathematica*)

$$x \qquad = (r_0 + v_0 t)\cos(\Omega t + \phi_0)\,, \quad y = (r_0 + v_0 t)\sin(\Omega t + \phi_0)\,,$$

-in the reference frame of the disk the trajectory is a spiral.

Summary

Today we discussed:

- Inverse problem of scattering
  - Analyzed how to solve various problems
- Fictitious forces in the non-inertial reference frame
  - Allow to get consistent description for free motion in the inertial and non-inertial reference frames